

JAMES494 RICHARDSON

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PROFESSIONAL SUMMARY

I'm a Machine Learning Engineer with 7 years of experience building and deploying production ML systems across fintech and healthcare sectors. I've spent the last 4 years leading technical initiatives that improved model accuracy and reduced inference latency by half. I'm passionate about mentoring junior engineers and translating complex problems into elegant, scalable solutions. I work well in fast-moving teams where I can contribute both code and architectural decisions.

TECHNICAL SKILLS

Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, neural networks, computer vision, NLP, recommendation systems

Data Engineering: Apache Spark, SQL, PostgreSQL, data pipelines, feature engineering, ETL processes

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform, Docker, Kubernetes

Programming Languages: Python, Java, SQL, Bash

Tools and Frameworks: Jupyter, Git, MLflow, Weights & Biases, Apache Airflow, Linux

Specializations: Model deployment, A/B testing, experimental design, time series forecasting

WORK EXPERIENCE

Senior Machine Learning Engineer

Synapse Analytics Limited

www.synapseanalytics.io

Bristol, United Kingdom

March 2022 - Present

Led development and deployment of a customer churn prediction model serving 50 million monthly predictions, reducing false positives by 34% through ensemble methods and careful threshold tuning

Architected a feature store using Apache Spark that reduced model training time from 8 hours to 45 minutes and improved reproducibility across 12 concurrent projects

Established MLOps best practices including model versioning, automated retraining pipelines, and monitoring dashboards that caught 3 model drift issues before they impacted users

Mentored 2 junior machine learning engineers, conducting weekly code reviews and leading 6 technical workshops on PyTorch optimization and production debugging

Machine Learning Engineer

Vertex Finance

www.vertexfinance.co

London, United Kingdom

July 2019 - February 2022

Built and maintained 8 gradient boosting models for real-time loan approval decisions that processed 2000 applications daily with 97% uptime

Improved model inference speed by 52% through quantization and ONNX conversion, enabling sub-100ms response times required by the business

Developed a comprehensive A/B testing framework that validated 15 model variants and prevented a risky deployment that would have decreased approval rates by 2%

Collaborated with data engineers to establish ETL pipelines using Apache Airflow that delivered clean training data to 6 downstream teams

Junior Machine Learning Engineer

HealthScope Technologies

www.healthscope.net

Manchester, United Kingdom

September 2017 - June 2019

Trained convolutional neural networks for medical image classification achieving 89% accuracy on X-ray pathology detection, supporting 12 radiologists in a pilot programme

Implemented data augmentation techniques that increased training dataset effective size by 400% and reduced overfitting on validation sets

Conducted hyperparameter tuning experiments across 200+ configurations using Bayesian optimization, documenting results in shared notebooks for team reuse

Supported production model serving infrastructure using TensorFlow and Docker, responding to 8 critical issues with same-day resolution

EDUCATION

MSc Computer Science (Machine Learning Specialization)

University of Manchester

Manchester, United Kingdom

2015 - 2017

BSc Computer Science

University of Bristol

Bristol, United Kingdom

2011 - 2015

GITHUB REPOSITORIES

<https://github.com/jrichardson494/time-series-forecaster>

Automated forecasting library using Prophet and LSTM networks with examples on energy consumption data

<https://github.com/jrichardson494/feature-store-framework>

Python implementation of a feature store with data versioning and backfill capabilities

<https://github.com/jrichardson494/neural-network-interpretability>

Tools for model explainability using SHAP and attention visualization for CNN models

<https://github.com/jrichardson494/recommender-system-benchmark>

Comparative analysis of collaborative filtering and content-based approaches with real-world datasets

CERTIFICATIONS

AWS Certified Machine Learning Specialty, Amazon Web Services, 2021

Machine Learning Engineering for Production Specialization, Coursera, 2020